



IILM UNIVERSITY

PYTHON PROGRAMMING

VIRTUAL INTERNSHIP — TASK SUBMISSION REPORT

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Enrollment Number: 25SCS1003000692

Program & Semester: B.Tech CSE (AIML), Semester 3

Organization: CodSoft (www.codsoft.in)

GitHub Repo: github.com/tanmaygiriinnovation/CODSOFT_TASKNO

INTERNSHIP OVERVIEW

CodSoft is a technology education and software development platform offering project-based virtual internships. As part of the Python Programming Internship, interns are required to build real-world console applications that reinforce core programming, logic-building, and problem-solving skills.

Tasks Completed

01

Password Generator

Generates strong, random passwords of user-defined length.

02

Rock-Paper-Scissors Game

An interactive game between user and computer with score tracking.

03

Contact Book

A console app to add, view, search, update and delete contacts.

TASK 1

Password Generator

A command-line tool that creates strong, random passwords based on a user-specified length, helping users adopt safer password practices.

Key Features

- Accepts desired password length as user input
- Combines uppercase, lowercase, digits and special symbols
- Uses Python's random / secrets module for randomness
- Prints the generated password directly to the console
- Can be re-run to instantly generate a new password

Tools & Concepts Used

- Python string module (ascii_letters, digits, punctuation)
- random.choice() for random character selection
- Loops and string concatenation
- Input validation for length
- Console-based I/O

TASK 1

Password Generator — Approach

1

Take Input

Ask the user to enter the desired password length.

2

Build Character Pool

Combine letters, digits, and symbols into one pool.

3

Generate Password

Randomly pick characters from the pool for the required length.

4

Display Output

Print the final generated password to the screen.

SAMPLE OUTPUT

```
Enter the desired password length: 12
```

```
Generated Password: xQ7!vB2#mZk9
```

```
Do you want to generate another password? (y/n): n
```

TASK 2

Rock-Paper-Scissors Game

An interactive console game where the user competes against the computer, with automatic result evaluation and optional score tracking across rounds.

Key Features

- User chooses Rock, Paper, or Scissors
- Computer makes a random selection each round
- Game logic decides the winner using standard rules
- Result of each round is displayed clearly
- Optional score tracking across multiple rounds
- 'Play Again' option to continue or exit

Tools & Concepts Used

- `random.choice()` for computer's move
- Conditional statements (if / elif / else)
- while loop for repeated rounds
- Variables for score tracking
- Clear, user-friendly console interface

TASK 2

Rock-Paper-Scissors — Approach

1

User Input

User selects rock, paper, or scissors.

2

Computer Move

System randomly picks its own move.

3

Decide Winner

Compare both choices using game rules.

4

Show Result & Score

Display outcome and updated score, ask to replay.

SAMPLE OUTPUT

```
Choose rock, paper, or scissors: paper
Computer chose: rock
You win! Paper covers Rock.
Score -> You: 1 | Computer: 0

Play again? (y/n): n
```

TASK 3

Contact Book

A menu-driven console application to store and manage personal contacts, supporting create, read, update, delete and search operations.

Key Features

- Stores name, phone number, email and address
- Add new contacts with full details
- View a list of all saved contacts
- Search contacts by name or phone number
- Update existing contact details
- Delete a contact from the list

Tools & Concepts Used

- Python dictionaries / lists for data storage
- Functions for add, view, search, update, delete
- Menu-driven loop for user interaction
- String matching for search functionality
- Clean, user-friendly interface design

TASK 3

Contact Book — Approach

1

Show Menu

Display Add / View / Search /
Update / Delete / Exit.

2

Take Choice

User selects an operation from the
menu.

3

Perform Operation

Corresponding function runs on the
contact list.

4

Loop / Exit

Return to menu until the user chooses
to exit.

SAMPLE OUTPUT

```
1. Add Contact 2. View Contacts 3. Search 4. Update 5. Delete 6. Exit
Enter your choice: 1
Name: Rohan Sharma | Phone: 9876543210
Contact added successfully!
```


INTERNSHIP COMPLETION CERTIFICATE

Issued by CodSoft on successful completion of the Python Programming Virtual Internship



CONCLUSION

Through this internship, I successfully completed 3 hands-on Python projects — a Password Generator, a Rock-Paper-Scissors Game, and a Contact Book. These tasks strengthened my understanding of Python fundamentals, logic building, and writing clean, user-friendly console applications, fulfilling the minimum requirement of the CodSoft Python Programming Internship.

THANK YOU

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